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Professor Kathleen M. Carley, Editor-in-Chief
Computational and Mathematical Organization Theory

Dear Professor Carley,

I submit “Code Geometry and Endogenous Enforcement Concentration in Codified Moral Systems: An Agent-Based Model” for consideration at *Computational and Mathematical Organization Theory*.

The paper asks how punishment authority in rule-governed organizations comes to be concentrated in a small, specialized minority, and why some such enforcement regimes stabilize into durable control while others fragment the organizations that host them. An agent-based model with symmetric agents and no pre-assigned enforcement roles, run across 3,246 simulations in five experimental specifications, yields four recurrent regimes: quiet, mixed enforcement, collapse, and capture. Enforcement activation is governed by a joint boundary between code legibility and enforcement reward. Once active, institutional delegation and compounding enforcement capital concentrate punishment in an endogenously selected minority; ablation shows that concentration depends on these two mechanisms rather than on preference asymmetry or architectural restrictions. Two theoretically plausible retention mechanisms (sanctified suffering and membership reward) fail to produce durable capture across their tested ranges, while endogenous degradation of the perceived outside option succeeds. The mechanism integrates delegation, path dependence, and compounding institutional capital into a single micro-level model, and the paper supplies a falsifiable scoring rubric for locating historically bounded regimes in the model’s parameter space.

The work falls within the journal’s interest in formal models of fundamental social processes, including destabilization, and in theories that explain the behavior of complex adaptive organizations. Religious and ideological systems serve as the empirical terrain because they provide the longest historical archive of codified enforcement, a domain the journal has previously addressed in models of radicalization and ethno-religious conflict; the mechanism itself is domain-general and applies to any organization that codifies conduct and attaches status to compliance.

One note on affiliation: my appointment is in computational chemistry, and this is independent research in computational social science conducted under the reproducibility standards of my primary field. The model is implemented in Python (Mesa); all code and data are public at <https://github.com/khatvangi/code-geometry-abm>, and the full results pipeline regenerates from committed data with a single script.

The manuscript is original, is not under consideration elsewhere, and has no competing interests or external funding to declare.

Sincerely,

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